

VERIDI LOGISTICS DELIVERY PERFORMANCE AUDIT

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AMALITECH DATA ANALYTICS CAPSTONE PROJECT

TOOLS USED:

- ❖ PYTHON (PANDAS, NUMPY, SEABORN AND MATPLOTLIB)
- ❖ GITHUB
- ❖ POWER BI (DAX)

PROBLEM STATEMENT

Veridi Logistics observed an increase in negative customer reviews and suspected that inaccurate delivery estimates and delayed deliveries were affecting customer satisfaction.

❖ Project Objectives

- ✓ Evaluate delivery performance.
- ✓ Identify regions with high late-delivery rates.
- ✓ Assess the impact of delivery delays on customer reviews.
- ✓ Develop an interactive dashboard to support business decision-making.

DATA PREPARATION & METHODOLOGY

❖ Dataset Used

- ✓ The dataset was downloaded from Kaggle
- ✓ and sourced from the Olist Brazilian E-Commerce Dataset.

❖ Data Sources

- ✓ The analysis utilized multiple relational tables, including Orders, Customers, Reviews, Products, and Product Category Translation datasets.
- ✓ Order_Items Dataset

DATA PREPARATION & METHODOLOGY

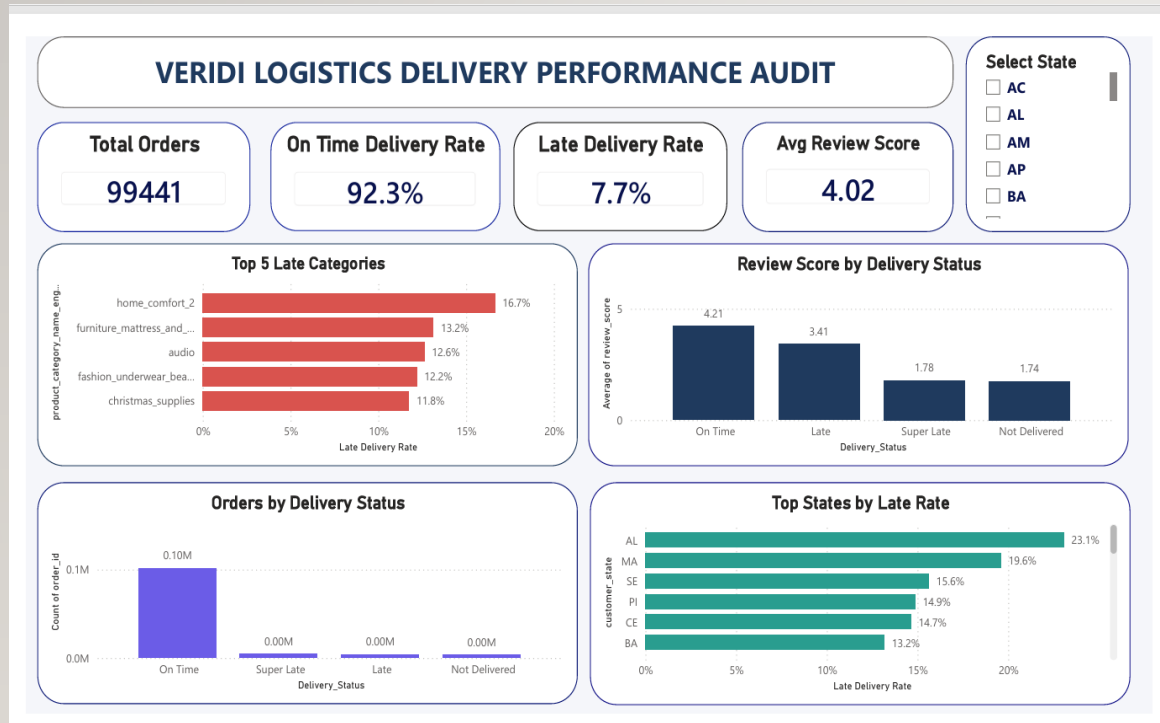
- **Data Preparation**

- Merged multiple datasets using order_id and customer_id.
- Handled missing values.
- Converted date columns to datetime format.
- Translated product categories from Portuguese to English.

- **Feature Engineering**

- Created:
- Days_Difference
- Delivery_Status
- Late_Flag
- Late Delivery Rate
- On-Time Delivery Rate

DASHBOARD OVERVIEW



- **Dashboard Components**
- Total Orders
- On-Time Delivery Rate
- Late Delivery Rate
- Average Review Score
- Top 5 Late Categories
- Top States by Late Rate
- Review Score by Delivery Status
- Orders by Delivery Status

KEY FINDINGS

- **Delivery Performance**

- Total Orders: 99,441
- On-Time Delivery Rate: 92.3%
- Late Delivery Rate: 7.7%

- **Geographic Analysis**

- Certain states recorded significantly higher late-delivery rates, indicating regional delivery challenges.

- **Product Categories**

- Home Comfort, Furniture & Mattress, and Audio products experienced the highest delay rates.



CUSTOMER SATISFACTION ANALYSIS

- **Review Scores by Delivery Status**

- Average Review Scores:

- On Time: 4.21

- Late: 3.41

- Super Late: 1.78

- Not Delivered: 1.74

- **Key Insight**

- Customer satisfaction decreases significantly as delivery performance worsens.

- This confirms that delivery delays are a major driver of negative customer reviews.



RECOMMENDATIONS & CONCLUSION

❖ Recommendations

1. Improve logistics planning for high-risk product categories.
2. Focus operational improvements on states with high delay rates.
3. Improve delivery monitoring and tracking systems.
4. Reduce delivery delays to improve customer satisfaction.

❖ Conclusion

- ✓ Veridi Logistics demonstrates strong overall delivery performance; however, targeted improvements in specific categories and regions can further reduce delays and improve customer experience.
- ✓ Thank You.

